Tutorial Exercise

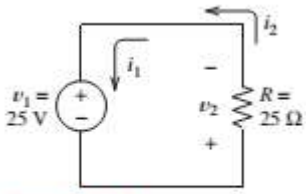


Figure 1.42 Circuit for Exercise 1.14.

Exercise 1.14 Analyze the circuit shown in Figure 1.42 to find the values of i_1, i_2, and v_2. Use the values found to compute the power for each element.

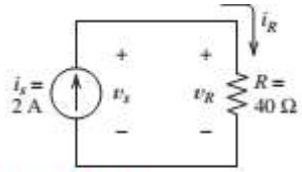


Figure 1.43 Circuit for Exercise 1.15.

Exercise 1.15 Figure 1.43 shows an independent current source connected across a resistance. Analyze to find the values of i_R, v_R, v_s, and the power for each element.

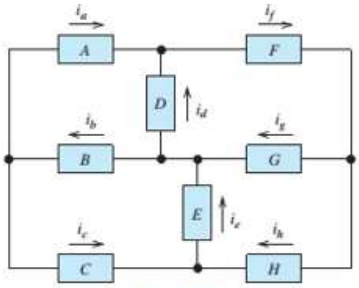


Figure P1.37

P1.38. Determine the values of the other currents in Figure P1.37, given that $i_a = 2$ A, $i_c = -3$ A, $i_g = 6$ A, and $i_h = 1$ A.

***P1.41.** Use KVL and KCL to solve for the labeled currents and voltages in Figure P1.41. Compute the power for each element and show that power is conserved (i.e., the algebraic sum of the powers is zero).

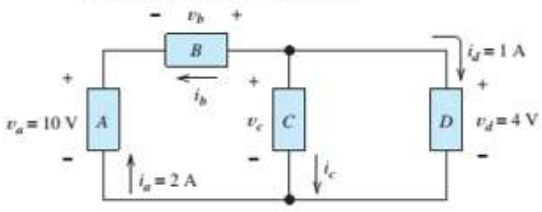


Figure P1.41

*P1.63. Consider the circuit shown in Figure P1.63. Use repeated applications of Ohm's law, KVL, and KCL to eventually find V_x.

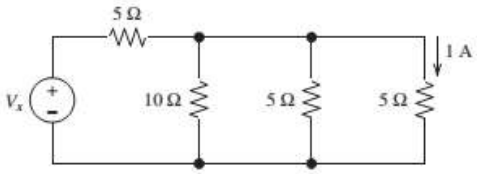


Figure P1.63

*P1.64. Consider the circuit shown in Figure P1.64. Find the current i_R flowing through the resistor. Find the power for each element in the circuit. Which elements are absorbing power?

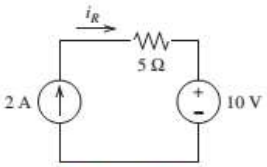


Figure P1.64